Akoizo Original Practice - 2027 Rotation Form 3

Division C | Chemistry Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) At constant temperature, a gas at 1.6 atm occupies 7 L. Compress it to 3.5 L. Find the final pressure.

Answer: _____
_____
_____

2. (2 points) At constant pressure, a gas occupies 7 L at 360 K. Find its volume at 720 K.

Answer: _____
_____
_____

3. (2 points) A rigid vessel contains a gas at 1.6 atm and 360 K. Find its pressure at 180 K with no gas lost.

Answer: _____
_____
_____

4. (2 points) Use PV=nRT with R=0.08206 L atm/(mol K). P=1.6 atm, V=7 L and T=360 K. Find the amount of gas.

Answer: _____
_____
_____

5. (2 points) A dry gas mixture has partial pressures 1.6 atm and 0.8 atm. Find total pressure.

Answer: _____
_____
_____

6. (2 points) A mixture has 5 mol oxygen and 15 mol nitrogen at total pressure 7 atm. Find oxygen partial pressure.

Answer: _____
_____
_____

7. (2 points) Gas collected over water has total pressure 103 kPa. Water-vapor pressure is supplied as 6 kPa. Find dry-gas pressure.

Answer: _____
_____
_____

8. (2 points) Two ideal gases at the same temperature have molar masses 4 and 16 g/mol. What is the diffusion-rate ratio light/heavy? Use Graham's law.

Answer: _____
_____
_____

9. (2 points) For 2 H2 + O2 -> 2 H2O, 5 mol O2 reacts completely with excess H2. How many moles of H2 are consumed?

Answer: _____
_____
_____

10. (2 points) Explain why gas-law temperature ratios must use kelvin rather than degrees Celsius.

Answer: _____
_____
_____

11. (2 points) [State/National] Determine the order with respect to A from the supplied initial-rate data.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

Answer: _____
_____
_____

12. (2 points) [State/National] Determine the order with respect to B from the supplied initial-rate data.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

Answer: _____
_____
_____

13. (2 points) [State/National] Calculate k from the initial-rate data. Give its value in M^-2 s^-1.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

Answer: _____
_____
_____

14. (2 points) [State/National] Predict the rate at [A]=0.48 M and [B]=0.10 M.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

Answer: _____
_____
_____

15. (2 points) [State/National] For a first-order reaction, k=0.05330769 s^-1. Use ln(2)=0.693 to calculate the half-life.

Answer: _____
_____
_____

16. (2 points) [State/National] A first-order reactant starts at 1.04 M. Find concentration after three half-lives.

Answer: _____
_____
_____

17. (2 points) [State/National] A zero-order reaction starts at 1.3 M and consumes reactant at 0.02 M/s. Find concentration after 10 s.

Answer: _____
_____
_____

18. (2 points) A catalyst increases rate without changing reactant/product free energies. Explain the mechanism and whether the equilibrium constant changes.

Answer: _____
_____
_____

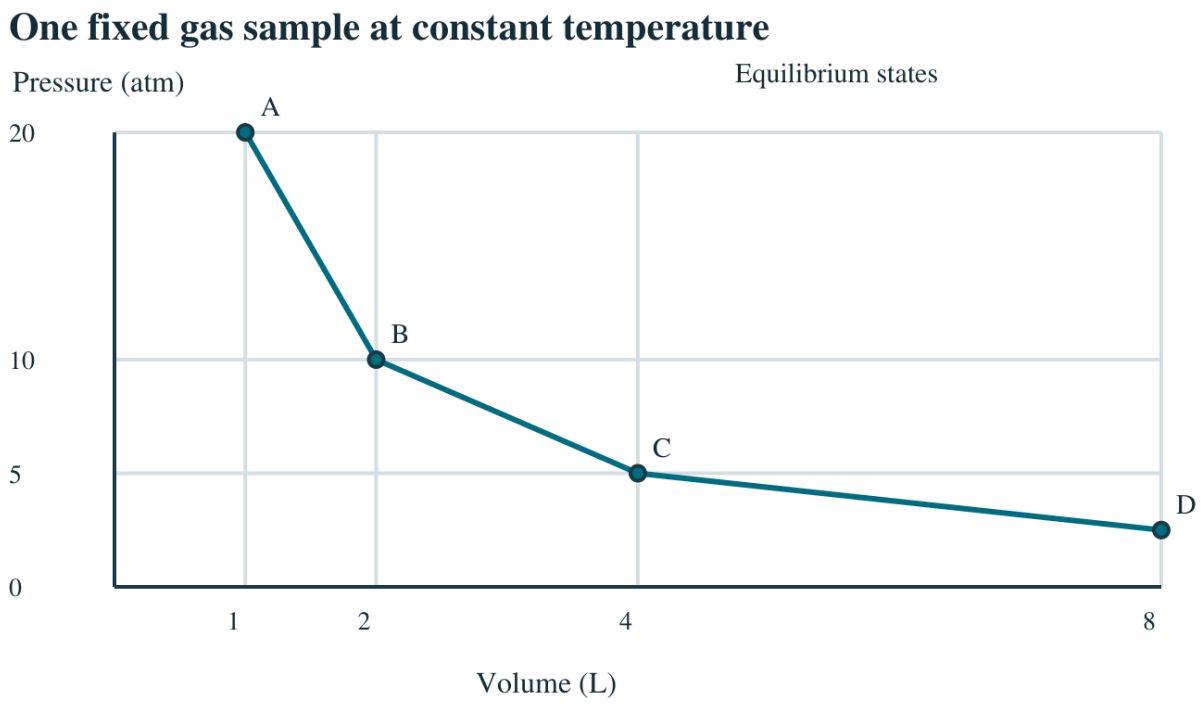
19. (2 points) Equal masses of the same solid react with excess acid under matched conditions. Why can smaller particles react faster?

Answer: _____
_____
_____

20. (2 points) [State/National] At 300 K a rate constant is k1; at 330 K it is 2k1. Using R=8.314 J/(mol K), ln(2)=0.693 and ln(k2/k1)=Ea/R(1/T1-1/T2), find Ea.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Read the pressure at state C.



F1. Ideal gas isotherm (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Calculate PV at B.

Refer to F1. Ideal gas isotherm (original synthetic teaching figure) shown with the first question in this figure group.

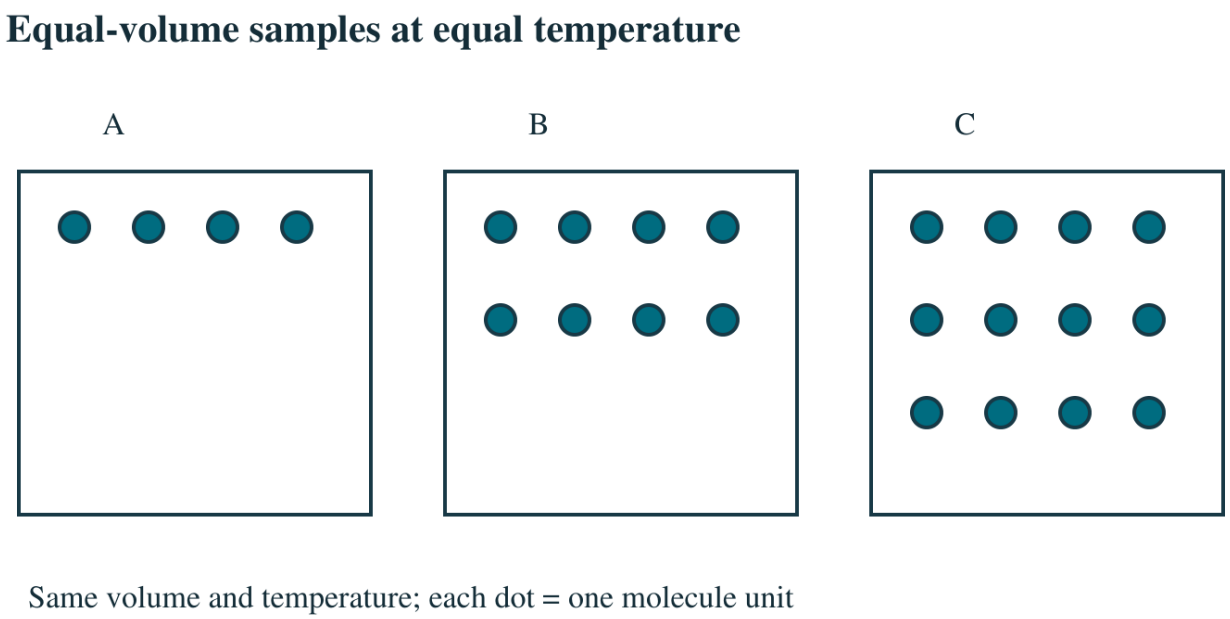
Answer: _____
_____
_____

23. (2 points) Use Figure F1. Use two plotted states to explain the volume-pressure relationship.

Refer to F1. Ideal gas isotherm (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Find the pressure ratio B/A.



F2. Ideal gas particle models (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. If A contains 0.10 mol, how many moles does C contain?

Refer to F2. Ideal gas particle models (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

26. (2 points) Use Figure F2. Would the dots justify a claim that C has faster molecules than A? Explain.

Refer to F2. Ideal gas particle models (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. [State/National] Determine reaction order with respect to A from runs 1 and 2.

Matched-temperature initial-rate measurements

Run	[A] (M)	[B] (M)	Rate (M/s)
1	0.10	0.10	0.005
2	0.20	0.10	0.020
3	0.10	0.20	0.010

F3. Initial-rate experiment (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. [State/National] Determine reaction order with respect to B from runs 1 and 3.

Refer to F3. Initial-rate experiment (original synthetic teaching figure) shown with the first question in this figure group.

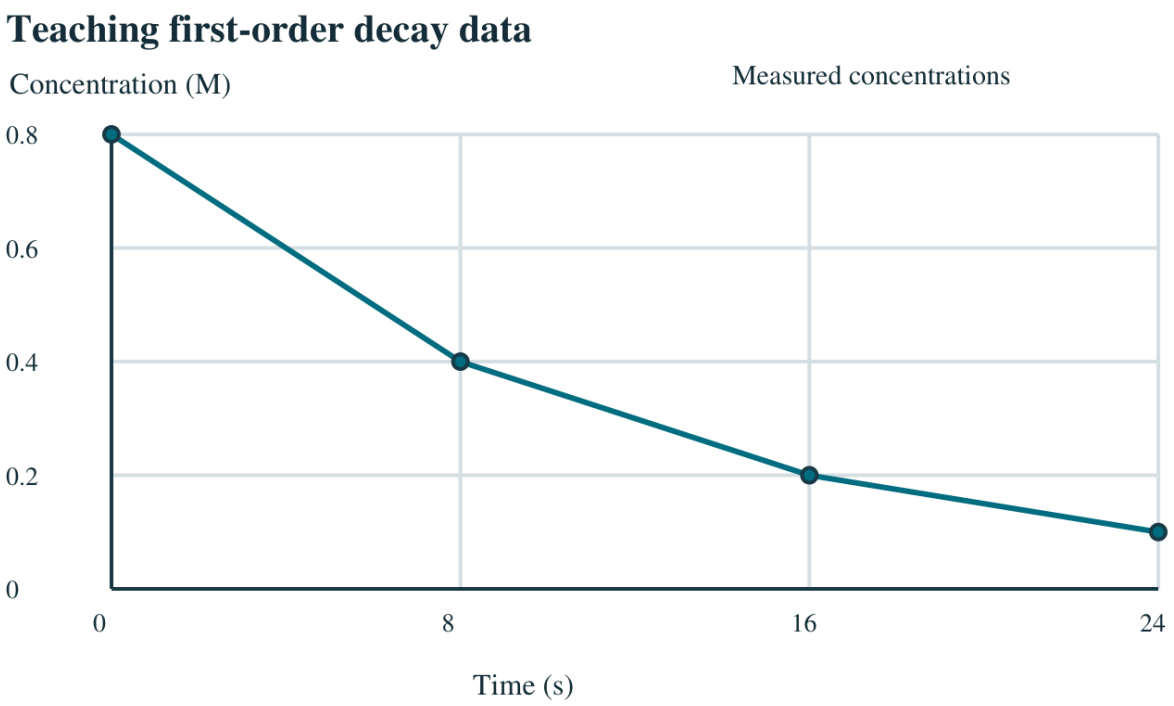
Answer: _____
_____
_____

29. (2 points) Use Figure F3. [State/National] Calculate k for rate=k[A]^2[B].

Refer to F3. Initial-rate experiment (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. [State/National] Read the half-life.



F4. Reactant decay curve (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. [State/National] Use k=ln(2)/half-life with ln(2)=0.693 to calculate k.

Refer to F4. Reactant decay curve (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. [State/National] Explain why the equal time intervals do not remove an equal concentration each time.

Refer to F4. Reactant decay curve (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____